Alertes Firmware : Messages d'Erreur BIOS/UEFI, Profils XMP et Pré-Boot

1. Contexte et Description Technique

Analyse des écrans d'alerte émis par le BIOS/UEFI (American Megatrends, Dell SupportAssist, HP SureStart, Apple Diagnostics) concernant la configuration de la RAM.

2. Symptômes Observés sur la Machine

Apparition d'un écran texte bloquant au démarrage : 'Overclocking failed! Please enter Setup', 'Memory PMU Training Error', ou code erreur PPM001.

3. Protocole de Diagnostic et Résolution

1. Entrer dans le BIOS (touche Suppr ou F2) et vérifier la fréquence actuelle par rapport au standard JEDEC (2133/2400 MHz en DDR4, 4800 MHz en DDR5).

2. Si le profil XMP/DOCP/EXPO échoue, augmenter légèrement la tension DRAM VDD (ex. de 1.35V à 1.38V) ou réduire la fréquence d'un palier.

4. Articles et Retours d'Expérience Extraits du Web

Extrait de [Memory Overclocking Fail (bios Cmos Reset Steps) - LaptopJudge] :
Memory Overclocking Fail (BIOS CMOS Reset Steps) When unstable RAM settings stop a PC from completing POST, clear the CMOS to restore safe firmware defaults. Shut down, unplug power, discharge the board, then use the correct CLR_CMOS pins or remove the CR2032 battery for about five minutes. Re-enter BIOS, disable XMP or DOCP, confirm JEDEC memory speed, and test stability before tuning again. Smart homes work because devices share clear rules and recover when one setting fails. A gaming PC needs the same discipline. If a memory profile pushes the system beyond what the processor, motherboard, or RAM can handle, the computer may freeze, restart, show POST code 55 or 00, or refuse to display an image. I treat a failed memory overclock as a firmware recovery task, not a Windows problem. The safest path is to restore default memory training, record a clean baseline, and only then investigate frame drops, thermal throttling, or input lag. Identifying Failed Memory Overclock Symptoms A failed memory overclock occurs when the motherboard cannot train or maintain the selected RAM speed and timings. “POST” means the power-on self-test that checks core hardware before Windows starts. A faile...

Extrait de [Ram Overclocking Failed Xmp (bios Voltage Adjustment)] :
RAM Overclocking Failed XMP (BIOS Voltage Adjustment) When an XMP profile crashes, the safest fix is controlled DRAM voltage tuning, not guesswork. Start from BIOS defaults, apply only the memory profile, and test stability. For DDR4, 1.35 V is common, with careful testing up to 1.40 V. Use MemTest86, check SPD timings, monitor temperatures, and return to JEDEC settings if errors continue. Establish a Clean Performance Baseline A baseline is a record of system behavior before any change. It should include memory speed, voltage, processor temperature, power draw, frame rate, and frame time. Without these figures, it is easy to mistake a driver problem or thermal throttle for unstable memory. A 60 FPS target equals a 16.7 millisecond frame time. At 144 FPS, each frame has only 6.9 milliseconds. A few delayed frames can therefore feel like input lag, even when the average frame rate looks acceptable. Before entering BIOS, record: - Current memory speed and timings - DRAM voltage shown by the motherboard - CPU and GPU temperatures during a 15-minute game session - Average FPS and 1% low FPS - Frame-time spikes using CapFrameX or similar logging software - CPU and GPU power in watts I a...

Sources Web Associées :

• Bios stuck at memory training - HP Support Community (https://h30434.www3.hp.com/t5/Desktop-Boot-and-Lockup/Bios-stuck-at-memory-training/td-p/9614287)

• Issue: Laptop became unable to boot after BIOS memory ... (https://h30434.www3.hp.com/t5/AI-PCs/Issue-Laptop-became-unable-to-boot-after-BIOS-memory/td-p/9708918)

• Memory Overclocking Fail (bios Cmos Reset Steps) - LaptopJudge (https://laptopjudge.com/memory-overclocking-fail/)

Catalogue Visuel : 10 Photos Réelles (07_bios_uefi_memory_alerts)

Ces 10 cas visuels issus du dataset illustrent les motifs caractéristiques de cette classe d'erreur :

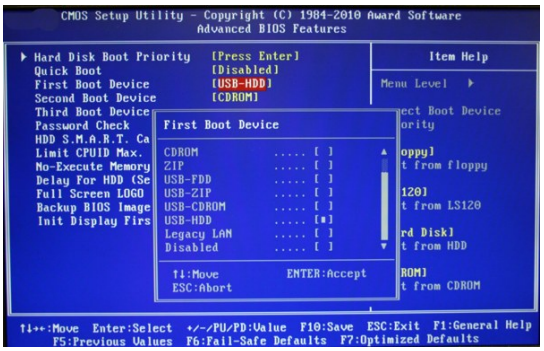


Figure 01 — 07_bios_uefi_memory_alerts (Échantillon #1)

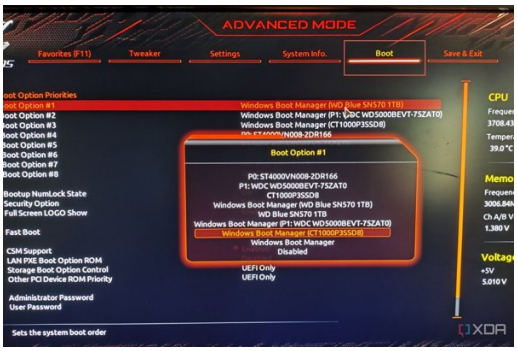


Figure 02 — 07_bios_uefi_memory_alerts (Échantillon #2)

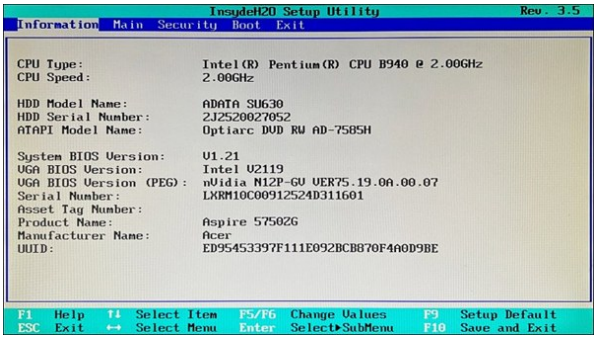


Figure 03 — 07_bios_uefi_memory_alerts (Échantillon #3)



Figure 04 — 07_bios_uefi_memory_alerts (Échantillon #4)

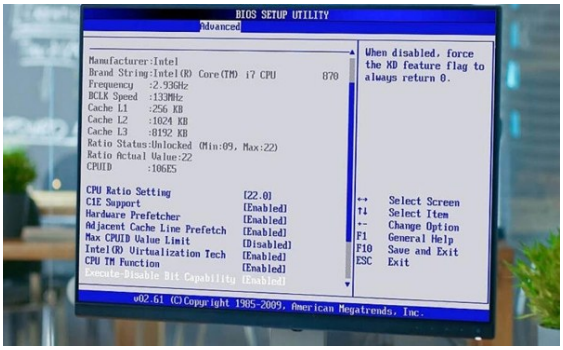


Figure 05 — 07_bios_uefi_memory_alerts (Échantillon #5)

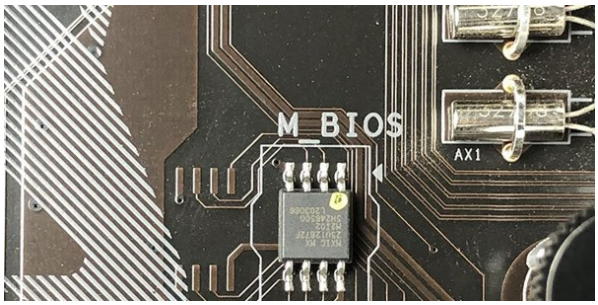


Figure 06 — 07_bios_uefi_memory_alerts (Échantillon #6)

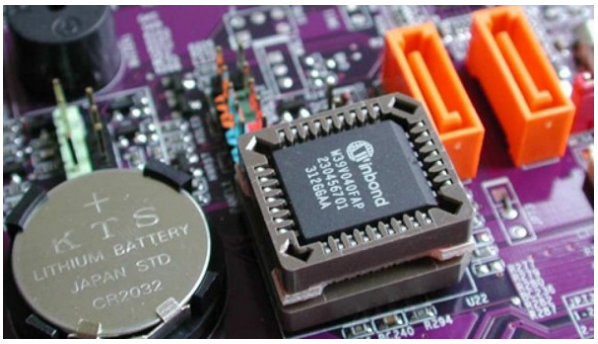


Figure 07 — 07_bios_uefi_memory_alerts (Échantillon #7)

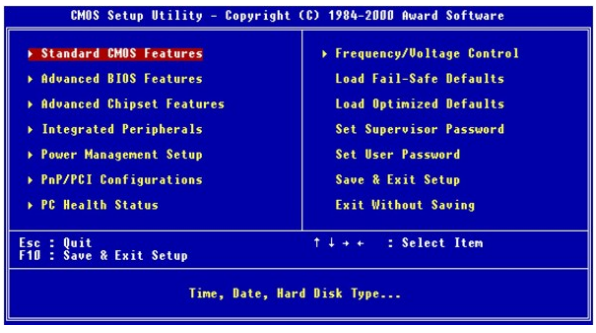


Figure 08 — 07_bios_uefi_memory_alerts (Échantillon #8)

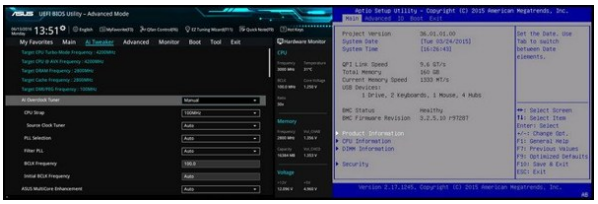


Figure 09 — 07_bios_uefi_memory_alerts (Échantillon #9)



Figure 10 — 07_bios_uefi_memory_alerts (Échantillon #10)